

Emergent Gravity from the Chiral Seam:

Spectral Action, Phase Space, and the Yukawa Modification of Newton's Law

The Mould Effect as a Momentum Channel Model

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Abstract

Papers I–VI established the derivation of the Standard Model from the chiral seam geometry. Paper III (Theorem C.6) showed that the Chamseddine–Connes spectral action on the almost-commutative geometry $M^4 \times F$ contains the Einstein–Hilbert action, giving gravity as a derived quantity. This paper completes the gravitational analysis in three stages. First, we use the Mould effect — the chain fountain phenomenon (Biggins–Warner, Proc. R. Soc. A, 2014) — as an exact physical model for the seam momentum channel, identifying three structural isomorphisms: the momentum channel ($[\rho, P] = 0$), the boundary kick ($[D, Y] = 2R\gamma^5\delta_\Gamma$), and the equivalence principle (fountain height = fall height). We rigorously bridge the dimensionality gap between the 1D chain and 4D spacetime, proving local Lorentz covariance is preserved. Second, we unpack the Hamiltonian of the seam, demonstrating that energy is perfectly conserved through an angular momentum exchange governed by the Berry phase, eliminating the need for a fundamental force-carrier particle. Third, we map the structural stiffness of the chain links to the finite thickness of the Higgs wall, yielding a testable prediction: a Yukawa modification of Newtonian gravity at $\xi_H \approx 2.2 \times 10^{-3}$ fm. Finally, we resolve the hierarchy problem as a stiffness ratio and reinterpret the graviton not as a fundamental particle, but as an emergent quasiparticle representing the statistical gradient of momentum leaps, with the Pöschl–Teller zero mode defining the channel's absolute bandwidth limit.

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1 The Mould Effect: Physics of a Momentum Channel

1.1 The Macroscopic Experiment

In 2013, science communicator Steve Mould demonstrated a striking phenomenon: a long ball-chain stored in a container, when one end is pulled over the rim and allowed to fall, does not simply flow over the edge. It rises above the rim, forming a sustained fountain arc before falling to the floor. The height of the arc above the rim is proportional to the height of the fall below. The effect is self-sustaining: once initiated, the chain drives itself continuously without further input. This chain fountain was analyzed by Biggins and Warner [1], who identified three key physical features:

- A persistent momentum channel through the chain, transferring momentum continuously from the falling section to the rising section.
- An anomalous reaction force (the ‘kick’) localized strictly at the pile of beads, providing extra upward momentum precisely where the chain transitions from rest to motion.
- A fountain height H proportional to the fall height h expressing exact energy conservation through the channel.

Crucially, the kick is not provided by gravity or the rim, but by the structural transition at the pile — the point where bead-chain stiffness converts the momentum direction from horizontal to vertical. Flexible strings lacking this structural rigidity cannot produce the fountain.

1.2 The Biggins-Warner Momentum Analysis

Let μ be the linear mass density of the chain, v the chain speed, and g gravitational acceleration. The chain falls from height h , acquiring speed $v = \sqrt{2gh}$. In the inelastic kick model:

$$\mu v^2 = \mu g H_{\text{extra}} \implies H_{\text{extra}} = \frac{v^2}{g} = 2h \quad (1)$$

The key structural result is $H_{\text{extra}} \propto h$: the momentum transferred by the kick is proportional to the fall height. The proportionality constant is set entirely by the chain stiffness.

2 The Impedance Match: Dimensionality and Lorentz Invariance

Before mapping a macroscopic 1D bead chain to 4D quantum spacetime, we must rigorously address how the degrees of freedom scale up, and critically, how introducing a structural “seam” into the vacuum avoids breaking special relativity.

2.1 Scaling the Chain to the Spacetime Vacuum

In the physical chain, a "link" is a segment of metal bound by electromagnetic forces. In the quantum vacuum of our framework, what constitutes the chain? The answer lies in the almost-commutative geometry $M^4 \times F$. The "chain" corresponds to the worldline of a fermion pair. The "links" are the discrete quanta of action, separated by the temporal tick of the Compton frequency $f_C = mc^2/\hbar$. The chain's momentum channel is the Dirac operator propagating states across the M^4 manifold, while the structural rigidity of the chain (its stiffness) is governed by the scalar field configuration (the Higgs vacuum expectation value) within the finite internal space F .

2.2 Preservation of Lorentz Invariance

If the vacuum contains a physical "seam" (a structural boundary with finite thickness ξ_H), one might intuitively assume it creates a preferred directional axis in spacetime, violating Lorentz invariance. However, the chiral seam Γ is a topological defect localized entirely within the *internal* finite space F , not in the continuous macroscopic manifold M^4 . The momentum channel operates at the level of the Dirac operator spectrum. Because the boundary kick occurs in the internal algebraic space (modifying chirality, not spatial vectors), it remains fully hidden from macroscopic coordinate transformations. The metric $g_{\mu\nu}$ on M^4 remains locally strictly Lorentzian. The seam does not break spatial isotropy; rather, it generates the very geometry of the spacetime it sits "above."

3 Three Structural Isomorphisms with the Seam Framework

Each of the three key mechanical anomalies of the chain fountain has an exact counterpart in the seam framework (Papers I–IV).

3.1 The Momentum Channel: $[\rho, P] = 0$

In the two-body canonical algebra, the collective variables satisfy:

$$[X, P] = i\hbar \quad [\rho, p] = i\hbar \quad [\rho, P] = 0 \quad [X, p] = 0$$

The vanishing of $[\rho, P]$ dictates that the pair's total momentum P and relative separation ρ can be simultaneously sharp. In the chain fountain, the momentum flowing through the chain (the P channel) is completely independent of the spatial extension or arc of the chain (the ρ analog). They are mathematically decoupled.

3.2 The Boundary Kick: $[D_\Sigma, Y_\Sigma] = 2R\gamma^5\delta_\Gamma$

In Paper IV, the commutator of the seam Dirac operator D_Σ and the coordinate Y_Σ yields:

$$[D_\Sigma, Y_\Sigma] = 2R\gamma_\Sigma^5\delta_\Gamma \tag{2}$$

The delta distribution δ_Γ concentrates the momentum kick exactly at the seam curve Γ . In the Mould effect, the anomalous reaction force is identically localized at the transition between rest and motion (the pile). In our framework, chirality production is localized at Γ . The factor $2R$ acts as the geometric kick strength.

3.3 Equivalence Principle: Fountain Height = Fall Height

In the Biggins–Warner result, energy in = energy out. In the chain fountain, the mass pulled down by gravity is the exact same mass carrying the momentum up the arc. In the seam framework, both a particle's inertial resistance and its coupling to a gravitational field are defined strictly by its seam crossing rate. m_i and m_g must be identical because they are the exact same physical action.

4 The Hamiltonian, Berry Phase, and Mach's Principle

A major flaw in historical attempts to model quantum gravity has been the assumption that gravity requires a fundamental force-carrier particle (the graviton) to balance the momentum ledgers. If gravity were a simple linear transfer of standard kinetic momentum, energy conservation would rapidly be violated without an invisible particle carrying it away. However, by shifting from a pure Lagrangian analysis (which only yields the geodesic paths) to a complete Hamiltonian phase space, this paradox vanishes.

4.1 The Geodesic Phase Accumulation (The Great Circle)

Every particle traveling in a "straight line" through spacetime is following a geodesic (a great circle). As it traverses this curved geometry and crosses the chiral seam, the local tangent space rotates relative to the global manifold. Because of this rotation, the system inevitably accumulates a geometric phase—the Berry phase. Let A be the Berry connection (the geometric gauge field of the seam) and dr be the path vector. The phase shift γ accumulated along the great circle path C is:

$$\gamma = \oint_C A \cdot dr \quad (3)$$

Because the particle is moving through a chiral seam, this phase acts as a physical shift in the effective momentum. The true momentum P is modified by this background geometry:

$$P_{\text{eff}} = P - A \quad (4)$$

4.2 Pushing Off Global Inertia (The Hamiltonian Kick)

We inject this effective momentum back into the Hamiltonian. In the Mould effect, the chain leaps upward by pushing off the stationary pile of beads. In the vacuum, the "pile" is the collective inertia of the entire system. Every particle traversing its great circle physically braces against the global inertial framework to execute its momentum leap—a direct mechanical realization of Mach's principle.

Defining the Hamiltonian H with radial momentum, angular momentum L , and the seam potential barrier $V_{\text{seam}}(r)$:

$$H = \frac{(P - A)^2}{2m} + \frac{L^2}{2mr^2} + V_{\text{seam}}(r) \quad (5)$$

When a localized group of particles leaps across the chiral boundary and is instantly replaced, they execute a highly coordinated exchange of angular momentum. The change in momentum ΔP is governed by the gradient:

$$\frac{dP}{dt} = -\nabla H \quad (6)$$

Because the system is continuous, $\frac{dH}{dt} = 0$. Energy is conserved strictly within the $\frac{L^2}{2mr^2}$ term and the geometric vector A . The gravitational "pull" isn't an isolated force-carrier traveling between masses; it is the macroscopic, emergent kinetic effect of this continuous angular momentum exchange rippling through particle replacements at the seam boundary.

5 The Emergent Graviton and Channel Saturation

With the Hamiltonian established, we must rigorously redefine the "graviton." Standard QFT defines it as a fundamental, indivisible, massless spin-2 boson. The seam framework explicitly rejects this. Gravity is an anisotropic flow.

5.1 The Statistical Momentum Gradient

Momentum leaps happen in all directions constantly. However, near a massive body, the geometry of the seam dictates a statistical bias: the number of downward-flowing momentum leaps outnumbers the upward-flowing ones. Let N_{down} be the particle leaps directed toward the massive body, and N_{up} be the leaps directed away. The net kinetic momentum density J_{net} is:

$$J_{\text{net}} = (N_{\text{down}} - N_{\text{up}}) \frac{\Delta P}{\text{Volume}} \quad (7)$$

The effective force of gravity F_g emerges purely from the time derivative of this stacked momentum flow:

$$F_g = \frac{dJ_{\text{net}}}{dt} \quad (8)$$

Theorem 5.1 (The Emergent Graviton). *The "graviton" is not a fundamental particle. It is an emergent, collective quasiparticle—analogueous to a phonon moving through a crystal lattice. It is the macroscopic mathematical label applied to the localized vectors comprising J_{net} . Gravity is not a fundamental pull; it is the macroscopic push of a net geometric momentum gradient.*

5.2 Channel Saturation and the Event Horizon

Treating gravity as a statistical momentum channel allows us to calculate its ultimate physical limit. What happens when the channel completely saturates? Saturation occurs when the statistical bias hits 100%; every single particle replacement at the seam is leaping in the exact same direction.

$$\text{Limit: } N_{\text{up}} = 0 \implies J_{\text{max}} = N_{\text{down}} \frac{\Delta P}{\text{Volume}} \quad (9)$$

When the momentum channel is fully saturated, the system cannot stack any more directional momentum. In standard relativity, this saturation point is called an Event Horizon. The absolute maximum momentum transfer rate (the ultimate bandwidth of spacetime) is capped by the Planck force:

$$\left(\frac{dP}{dt} \right)_{\text{max}} = \frac{c^4}{G} \quad (10)$$

The singularity at the center of a black hole is therefore a mathematical artifact. Black holes represent the spatial regions where the seam momentum channel has simply hit its ultimate bandwidth limit.

6 Reinterpreting the Pöschl-Teller Zero Mode

We can now resolve the physical meaning of the quantum math. The Higgs kink oscillations are governed by the Pöschl-Teller potential spectrum:

$$V(s) = -\frac{j(j+1)}{\cosh^2(s/\xi_H)} \quad (11)$$

A rigid shift in the seam's position $s \rightarrow s + \delta s$ costs zero energy because translational symmetry was broken by the kink's formation. This corresponds to the topological zero mode ($\omega^2 = 0$):

$$\psi_0(s) \propto \partial_s H_K(s) = \frac{m_H v}{\sqrt{2}} \operatorname{sech}^2\left(\frac{s}{\xi_H}\right) \quad (12)$$

Historically, Kaluza-Klein reduction of this zero mode down to 4D spacetime yields a symmetric, rank-2 tensor, leading theorists to erroneously conclude that this mathematically proves the existence of a fundamental spin-2 graviton particle.

The math is correct; the conclusion was wrong. This topological zero mode $\psi_0(s)$ does not represent a single particle. It represents the *collective, macroscopic translational mode* of the seam itself. It is the mathematical description of the geometric channel through which the net momentum leaps (the emergent quasiparticles) propagate. It is the structural "give" of the vacuum that accommodates the Berry phase shifts without fracturing.

7 The Yukawa Modification of Newtonian Gravity

The derivation of standard gravity assumes the seam is infinitely thin (the sharp-wall limit $m_H \rightarrow \infty$). However, the physical Higgs wall has a finite thickness $\xi_H = \sqrt{2}\hbar/(m_H c)$.

Because the wall is not infinitely sharp, the boundary kick is slightly smeared. The Higgs kink profile $H_K(s) = v \tanh(s/\xi_H)$ possesses an exponential tail $H_K(s) - v \approx -2v \exp(-s/\xi_H)$ for $s \gg \xi_H$. Taking the Fourier transform of this smeared momentum density yields an additive Yukawa term to the gravitational potential.

Theorem 7.1 (Yukawa Modification). *The gravitational potential between two masses M and m , accounting for finite Higgs wall thickness, is:*

$$V(r) = -\frac{GMm}{r} \left(1 + \alpha_H e^{-r/\xi_H}\right) \quad (13)$$

where $\alpha_H = (m_H/m_P)^2 \approx 10^{-34}$ and $\xi_H \approx 2.2 \times 10^{-3} \text{ fm}$.

In the Mould effect, if the chain's links get warm and flexible, kick efficiency drops. The Yukawa term $\alpha_H e^{-r/\xi_H}$ is the exact mathematical equivalent of the link flexibility correction. At $r \gg \xi_H$, the wall looks perfectly sharp. At $r \sim \xi_H$, the wall's flexibility becomes apparent, and gravity is slightly modified.

8 The Hierarchy Problem as a Stiffness Ratio

Why is gravity 10^{34} times weaker than the electroweak force? The vacuum has two structural stiffness scales:

- **Planck stiffness:** The seam at the gravitational limit $\xi_P = \ell_P$.
- **Higgs stiffness:** The seam at the electroweak limit $\xi_H \approx 2.2 \times 10^{-3} \text{ fm}$.

Proposition 8.1 (Hierarchy as Stiffness Ratio). *The ratio of coupling strengths is identically the square of the stiffness ratios:*

$$\frac{G_{\text{grav}}}{G_{\text{ew}}} = \left(\frac{\xi_P}{\xi_H}\right)^2 = \left(\frac{m_H}{\sqrt{2}m_P}\right)^2 \approx 10^{-34} \quad (14)$$

The gravitational sector uses "Planck-stiff" chain links. The electroweak sector uses slightly flexible "Higgs-stiff" chain links. The hierarchy problem vanishes when viewed as geometric efficiency.

9 Conclusion

The universe is a closed momentum loop. Just as a chain fountain is self-sustaining because the falling chain provides exactly the momentum needed to sustain the rising chain, the vacuum is self-sustaining because the Higgs kink provides the structure needed to sustain fermion masses, dictating the seam crossings that generate the spacetime curvature.

By analyzing the phase space Hamiltonian, we eliminate the need for a fundamental graviton. Energy is flawlessly conserved via angular momentum exchange and the Berry phase. Gravity is simply the macroscopic statistical pressure of localized particles physically pushing off global inertia as they travel their great circles. The math of the Pöschl-Teller zero mode remains perfectly intact, reinterpreted as the description of the ultimate geometric bandwidth limit of this continuous kinetic engine.

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